

What makes a good prediction?

K Arnold

Objectives

- Compare and contrast regression tasks and classification tasks, and give examples of each
- Identify two different ways of measuring accuracy for regression and for classification
- Identify several reasons why a model may predict better on some subsets of data than others

Types of Tasks

- **regression**: predict a *number* ("continuous")
 - number should be "close" in some sense to the correct number
- **classification**: predict a *category*
 - which one of these two groups? three groups? 500,000 groups?
 - could ask: "how likely is it to be in group i "

Examples of regression tasks?

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- How much time will this person spend watching this video?
- How much will it rain in GR next year?
- How big a fruit will this plant produce?
- How much will this home sell for?

...

Running regression problem: housing prices in Ames, Iowa. Details:

- Paper
- Data Dictionary

Examples of classification tasks?

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- Which word did this person mean to type?
- Will this person "Like" this post?
- Is this a picture of the inside or outside of the restaurant?
- Is this person having a seizure?

Running example: *seizure classification*.

First FDA-approved AI-powered medical device: Empatica **Embrace2**, company founded by MIT data scientist Rosalind Picard



What makes a good prediction? *Regression*

For a single observation:

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- **absolute error**
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Across the entire dataset:

- **average error**: do we tend to predict too high? too low? "*bias*"
- **max** absolute error
- **mean** absolute error
- **mean squared error** (MSE)
- normalized squared error: $\text{MSE} / \text{Variance}$
 - The confusingly named "R-two" = $1 - \text{normalized squared error}$

What makes a good prediction? *Classification*

	Seizure happens	No seizure happens
Seizure predicted	True positive	False positive (Type 1 error)
No seizure predicted	False negative (Type 2 error)	True negative

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- **Accuracy** (% correct) = $(TP + TN) / (\# \text{ episodes})$
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- **Sensitivity** ("true positive rate") = $TP / (\# \text{ actual seizures})$
 - Sensitivity = $1 - \text{False negative rate}$
- **Specificity** ("true negative rate") = $TN / (\# \text{ actual seizures})$
 - Specificity = $1 - \text{False positive rate}$
- [Wikipedia article](#)

If you were designing a seizure alert system, would you want sensitivity and specificity to be high or low? What are the trade-offs associated with each decision?

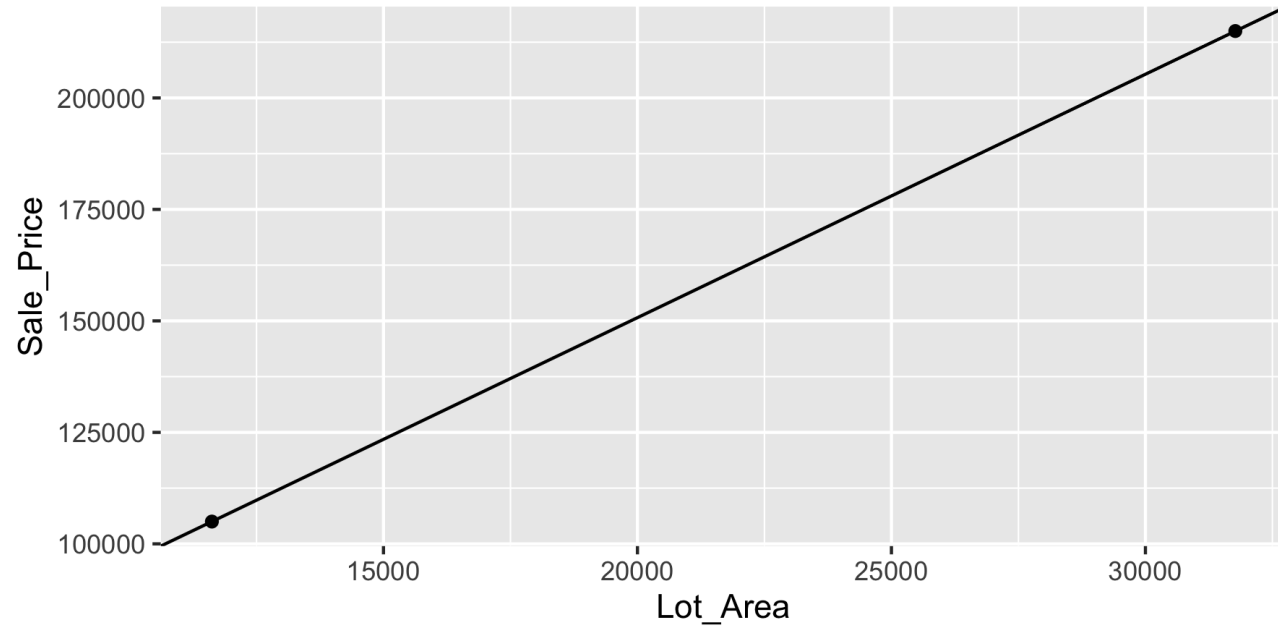
Validation

Key point: you *must* evaluate predictions on *unseen* data

Hey look! I can exactly predict how much a home will sell for!

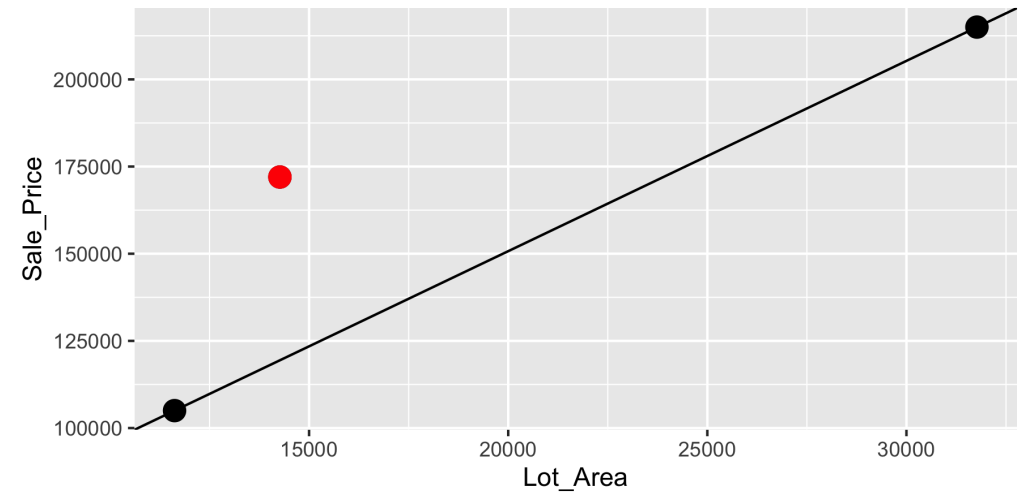
Lot_Area	Sale_Price
31770	215000
11622	105000

sale price = 41548.54 + 5.459599 * lot area



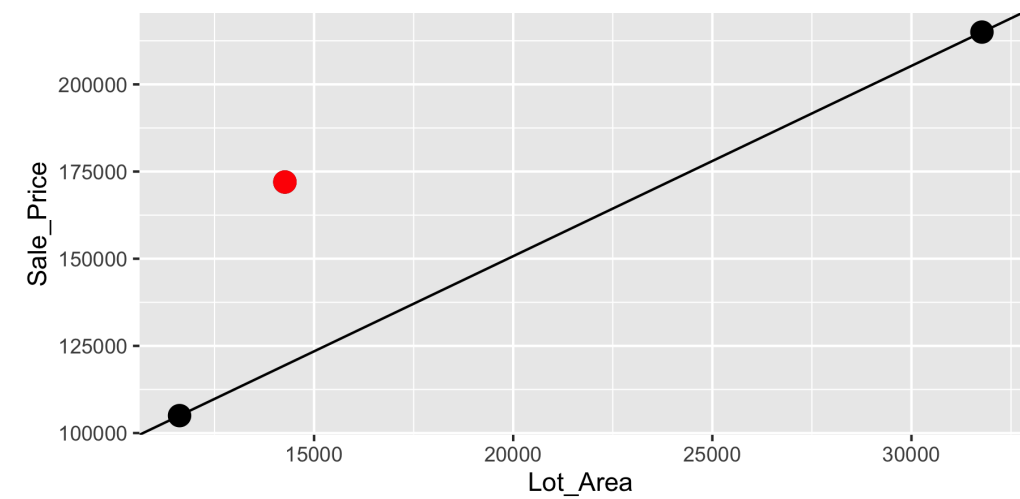
Validation: *unseen* data

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11622	105000
14267	172000



Lot_Area	Sale_Price	predicted	residual
31770	215000	215000.0	0.00
11622	105000	105000.0	0.00
14267	172000	119440.6	52559.36

Oh ok, I'll just fix that one...

Lot_Area	Bsmt_Unf_SF	Sale_Price
31770	441	215000
11622	270	105000
14267	406	172000

sale price = -37769.46 + 1.5311432 * lot area + **462.8685748 * basement sq ft**

and look, it works!

Lot_Area	Bsmt_Unf_SF	Sale_Price	predicted	residual
31770	441	215000	215000	0
11622	270	105000	105000	0
14267	406	172000	172000	0

Do you really think so?